DIGITAL TWIN · EAPH AIR HEATER CHECK · 4 NOV 2026

Does the twin's EAPH follow the plant?

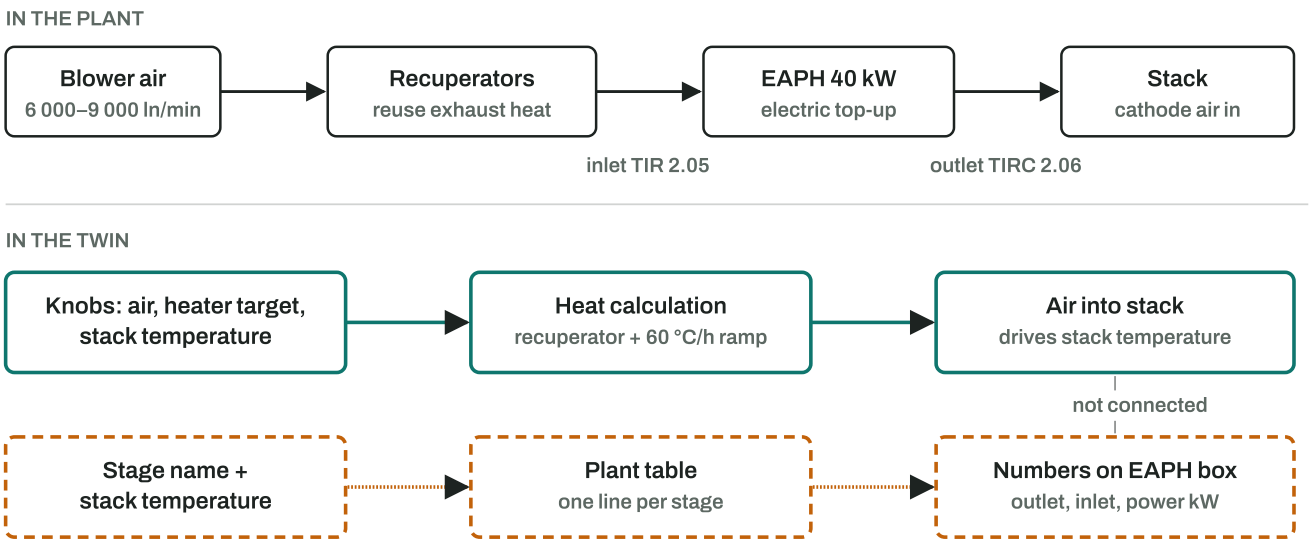
Update 4 Nov: the changes are made (twin v17). This page and pages 2–4 describe the twin *before* the change; results after are on Changes made.

> **Partly. The air temperature the twin calculates behind the heater follows the plant within a few degrees. The numbers printed on the EAPH box (outlet temperature and power) come from a separate table, so they are right on average but do not follow the plant minute by minute.**

Why it matters: the calculated air temperature is what warms the stack in the twin, and that part works. The printed power is what an operator would read, and it stays flat while the real heater goes up and down.

What the EAPH is and how the twin handles it

The EAPH (electric air pre-heater) is a 40 kW electric heater. Air from the blower first passes two recuperators, heat exchangers that reuse exhaust heat. The EAPH then tops it up to the target temperature before it enters the stack, like a hair-dryer at the end of an already warm pipe.



Top: the plant air path. Bottom: the twin makes its EAPH numbers in two separate ways. The green path is calculated and drives the stack. The orange dashed path is read from a table of plant data, and its numbers are what the EAPH box shows. The two paths do not talk to each other.

Scorecard

What the twin gives	Where	Miss vs plant	Verdict
Air into the stack (calculated)	Heat-up, hot standby	2 °C or less	follows
Air into the stack (calculated)	Fuel switch, part load	5–8 °C	follows
Air into the stack (calculated)	Air-flow cooling (heater off)	+39 °C too warm	off
Outlet number shown on EAPH box	Heat-up air / N_2+H_2	28 / 18 °C	close
Power number shown on EAPH box	Stage average	within 0.7 kW (fuel switch -5.4)	close
Power number shown on EAPH box	Ups and downs inside a stage	match 0.08 (1 = perfect)	does not follow

Test: 8 real plant periods (97 hours; 2 cold starts, 3 runs from hot standby through fuel switch to part load, 2 air-flow cool-downs, 1 shutdown). The plant's own knob settings were fed into today's twin (v16) every minute and the twin's EAPH values were compared with the plant's. The table in the orange path was built from the same plant days, so its average is right by construction. Fuel switch and part load cover only 53 and 99 minutes.

Air temperature →

The calculated value follows the plant. The printed value can be 18–27 °C away from it.

Heater power →

The printed power is one fixed number per stage. A calculated version would follow the plant's ups and downs.

What to change →

The four changes that were proposed.

Changes made →

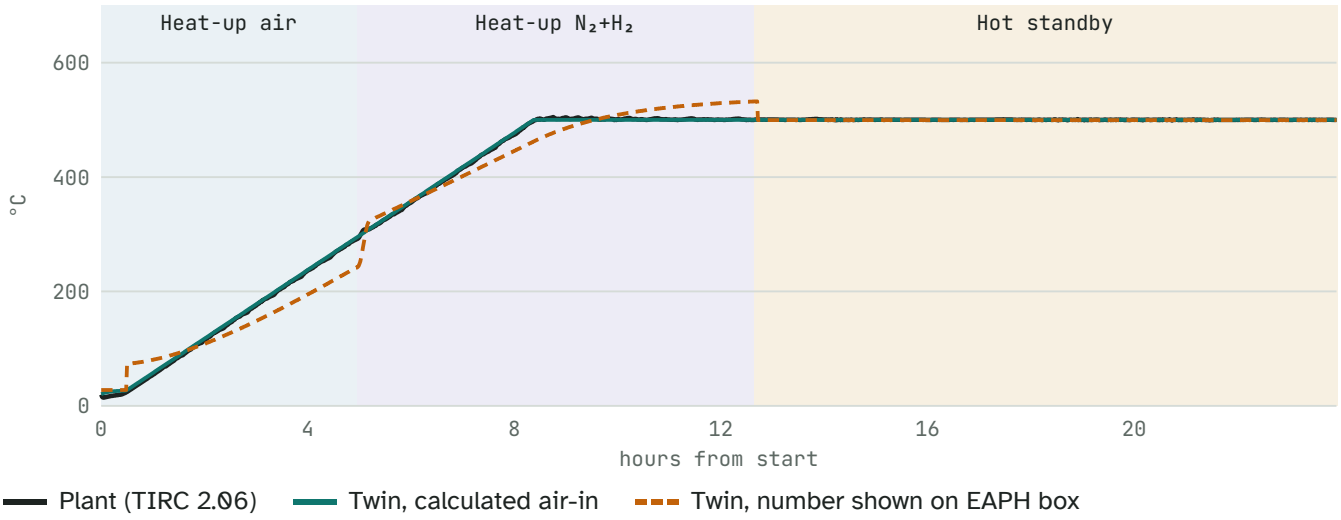
Done 4 Nov: the power now follows the plant.

IF YOU REMEMBER ONE THING

The twin heats the stack with the right air temperature, but the EAPH box shows table values, so its power reading stays flat while the real heater moves.

Air temperature: the calculation follows, the printed number lags

> **Inside the twin, the air leaving the EAPH warms at the plant's own pace and stops at the same target. The number printed on the EAPH box is looked up from a table instead, and during heat-up it sits 18–27 °C away from the twin's own value.**



Cold start, 24 hours (one of the 8 test periods). Plant outlet TIRC 2.06 in black; the twin's calculated air into the stack in green lies on top of it. The orange dashed line is the number the EAPH box shows: it follows the stack temperature through a table, so it rises in steps and runs ahead of or behind the real air. Shaded bands are the plant stages.

Per stage, all 8 test periods

Stage	Minutes	Plant outlet °C (median)	Calculated: average miss °C		Printed: average miss °C	Printed vs calculated °C
Heat-up air	575	147	+2	follows	28	27
Heat-up N_2+H_2	929	500	-0	follows	18	18
Hot standby	1302	500	-0	follows	2	2
Fuel switch	53	585	-5	follows	11	7

Stage	Minutes	Plant outlet °C (median)	Calculated: average miss °C		Printed: average miss °C	Printed vs calculated °C
Part load	99	608	-8	follows	16	18
Air-flow cooling	2355	245	+39	off	37	7
Shutdown	530	195	-15	close	5	20

Why the calculation works: the twin copies the plant's habits. The heater raises its outlet by 60 °C per hour, as the plant did in 9 of 12 heat-ups, and holds the target exactly (500 °C in hot standby, 580 in fuel switch, 600 in part load).

Where it fails: in **air-flow cooling** the heater is off, and the twin's recuperator guess makes the air about 39 °C too warm. This is the known recuperator point from the earlier twin-vs-plant check (decision D there), not an EAPH fault. The printed value happens to be closer there (37 °C) because it is copied from plant data.

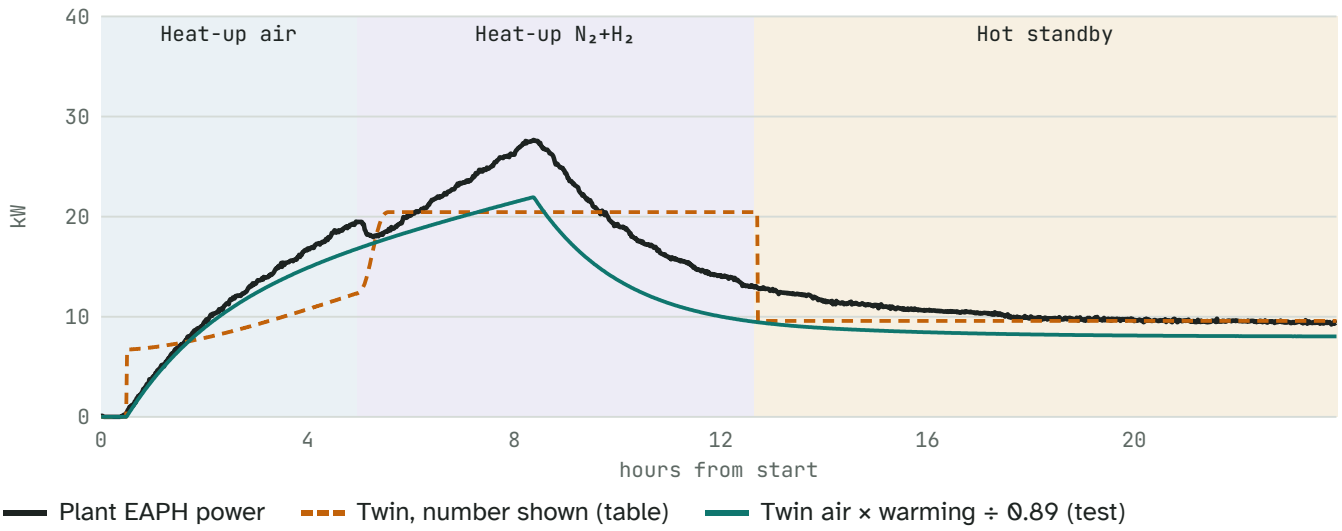
Fuel switch and part load: only 53 and 99 minutes in the test, so treat those two rows as a first look.

IF YOU REMEMBER ONE THING

The twin's own air temperature is right; the problem is only that the EAPH box prints a different, table-based number.

Heater power: right on average, flat inside each stage

The twin prints one fixed power per stage, the plant's median. The real heater moves a lot inside a stage: 14 to 26 kW while heating with N_2 + H_2. Inside a stage the twin's printed power matches those moves with a score of 0.08, where 1 is a perfect match.



Same cold start as page 2. Black: plant EAPH power. Orange dashed: the twin's printed power, a flat step per stage. Green: a test version the twin does not have yet, calculated as air flow × warming across the heater ÷ 0.89. The plant data show that 89–91 % of the electric power ends up in the air. The green line rises and falls with the plant.

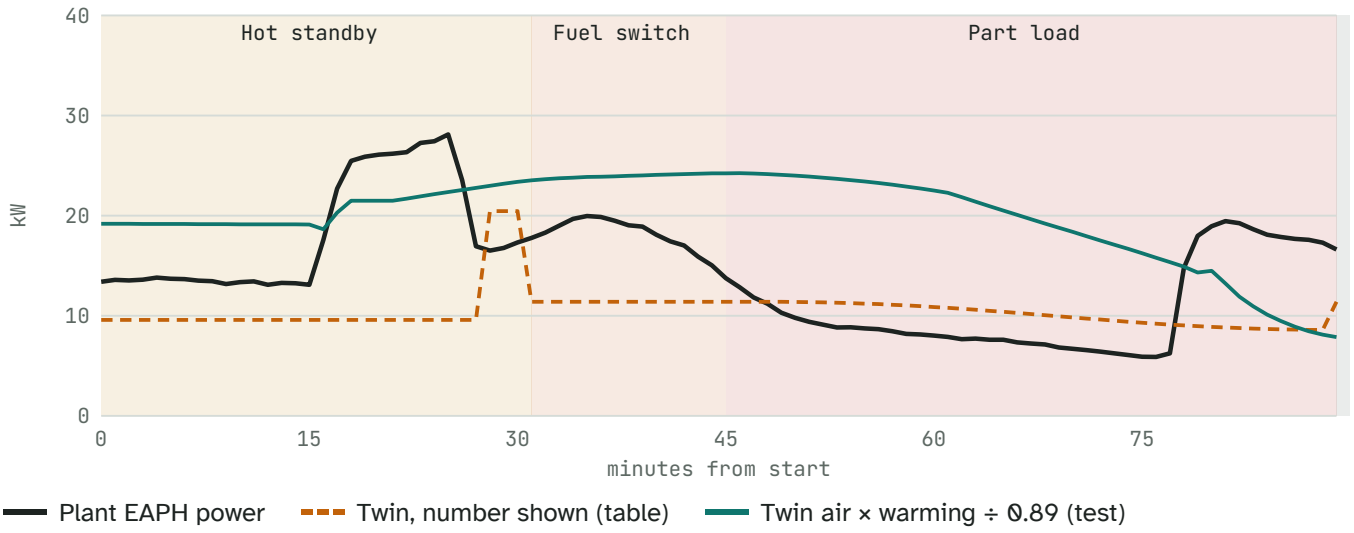
Per stage, all 8 test periods (kW)

Stage	Plant median	Plant usual range (10–90 %)	Twin printed	Printed: average miss	Calculated test	Calculated: average miss
Heat-up air	11.9	2–18	8.5	3.8	11.1	1.1
Heat-up N_2+H_2	20.0	14–26	20.4	3.7	17.3	4.1
Hot standby	10.1	10–12	9.6	1.3	8.3	2.3
Fuel switch	15.2	7–19	9.8	5.1	23.5	7.4

Stage	Plant median	Plant usual range (10–90 %)	Twin printed	Printed: average miss	Calculated test	Calculated: average miss
Part load	9.2	5–21	9.9	5.7	19.2	8.6
Air-flow cooling	0.1	0–0	0.0	0.1	0.0	0.1

Over the whole test both have about the same average miss (printed 1.4 kW, calculated 1.5 kW). The difference is in the shape: inside a stage the calculated one follows the plant (0.94) and the printed one does not (0.08), measured over 9 stage runs where the plant power moved.

Why the calculated version overshoots with gas on



Hot standby → fuel switch → part load (90 minutes). With gas burning, the green test line stays at 20–24 kW while the plant drops to 7–10 kW. The plant's jump to 28 kW in hot standby is the operator raising the heater target before opening the gas; the printed twin value (orange) does not see it.

The reason is the air arriving at the heater. With gas on, the exhaust burner runs hotter, so the recuperators deliver warmer air and the EAPH has less to add. The twin's recuperator does not know about the gas yet, so it thinks the heater has more work to do:

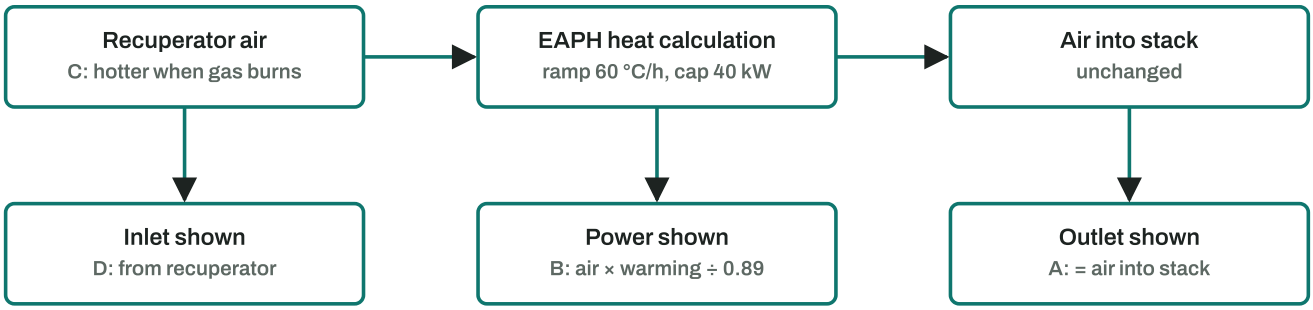
Stage	Plant air reaching EAPH °C (TIR 2.05)	Twin recuperator guess °C	Twin minus plant
Heat-up N_2+H_2	334	376	+42
Hot standby	433	447	+13
Fuel switch	529	493	-35
Part load	589	516	-73

IF YOU REMEMBER ONE THING

Printed EAPH power is a stage average. To make it move like the plant, the twin must calculate it, and first learn that the recuperator air is hotter when gas is on.

Making the EAPH box tell the same story as the calculation

> **The heat calculation already follows the plant. The fix is to let the EAPH box read from that calculation instead of from the table. Before the power can follow the plant in fuel switch and part load, the recuperator must learn that gas makes its air hotter.**



Proposed wiring. Every EAPH number comes from one green calculation. The letters match the decisions below. All four were made on 4 Nov; the built version differs in one place (see Changes made).

Decisions (all taken 4 Nov: C → A → B → D)

A · Print the twin's own air temperature on the EAPH box

Small change. Removes the 18–27 °C gap between the printed and calculated values during heat-up. In air-flow cooling the printed value would get worse (+39 °C) until the recuperator is fixed.

B · Calculate EAPH power from air flow × warming ÷ 0.89

Power then rises and falls like the plant (inside-stage match 0.94 instead of 0.08). Needs C first, or fuel switch and part load show about 10 kW too much.

C · Make the recuperator air hotter when gas is burning

Plant air reaches the EAPH 35–73 °C warmer than the twin thinks in fuel switch and part load. To be

D · Inlet temperature TIR 2.05 from the recuperator calculation

Keeps all three EAPH numbers on one calculation. Today the printed inlet misses by 25 °C on average.

fitted on the plant days, then checked again on these 8 periods.

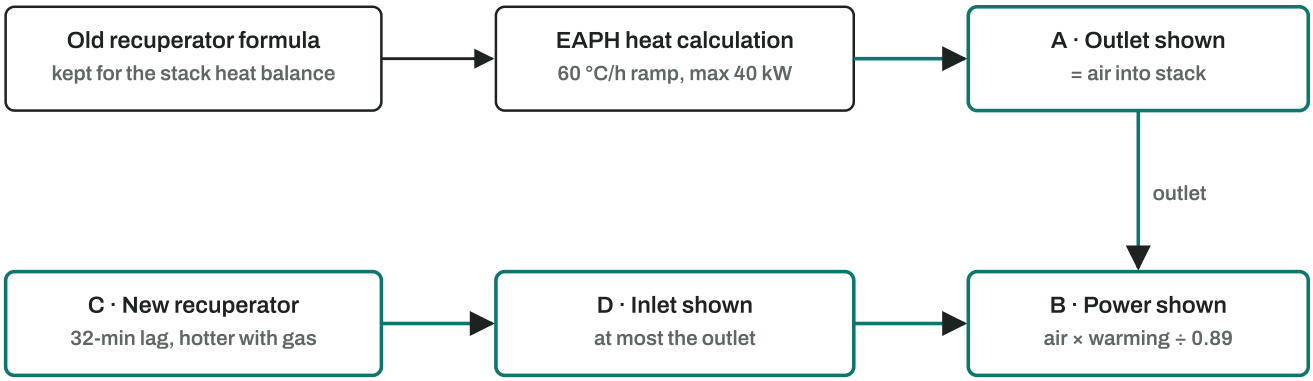
Done in this order and re-tested on the same 8 periods: results on Changes made.

IF YOU REMEMBER ONE THING

One calculation should feed every EAPH number; first teach the recuperator that gas makes its air hotter.

All four changes are in. The EAPH power now moves like the plant.

> **The order was C, A, B, D. The EAPH box now reads from the model. The power misses the plant by 0.66 kW per minute instead of 1.42. Inside a stage it follows the plant's ups and downs (match 0.95 instead of 0.08). The stack temperature did not change at all.**

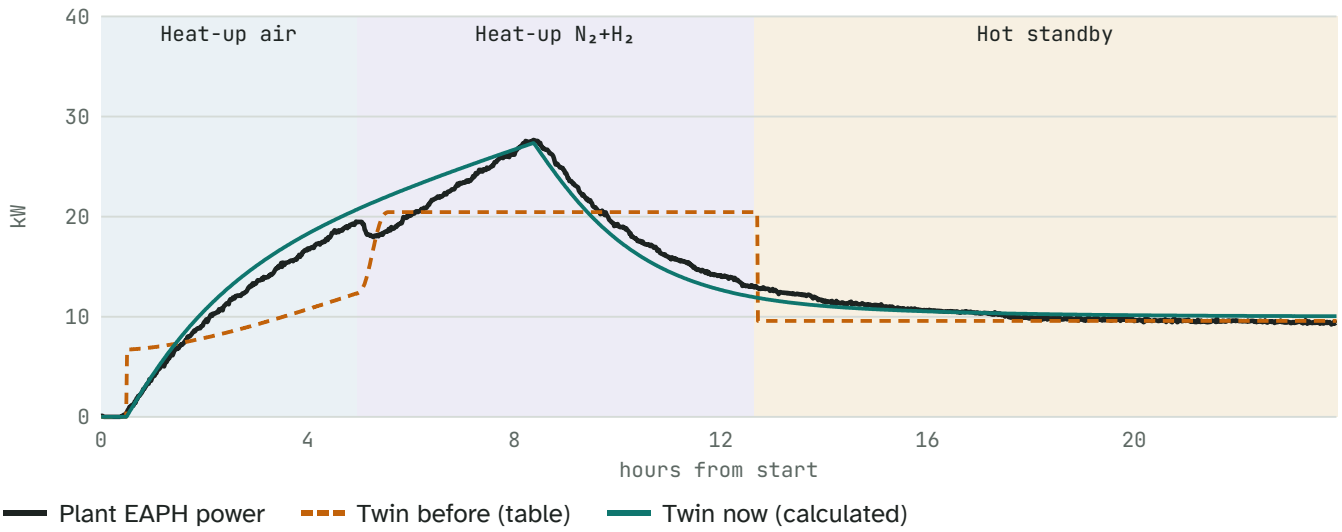


As built. Green = new model readings on the EAPH box. One change from the plan on page 4: the stack heat balance keeps its old recuperator formula. With the new, slower recuperator inside it, the air stayed warm while the stack cooled, and the 9000 ln/min cool-down missed by 156 °C instead of 74. So the new recuperator feeds only the inlet and power readings.

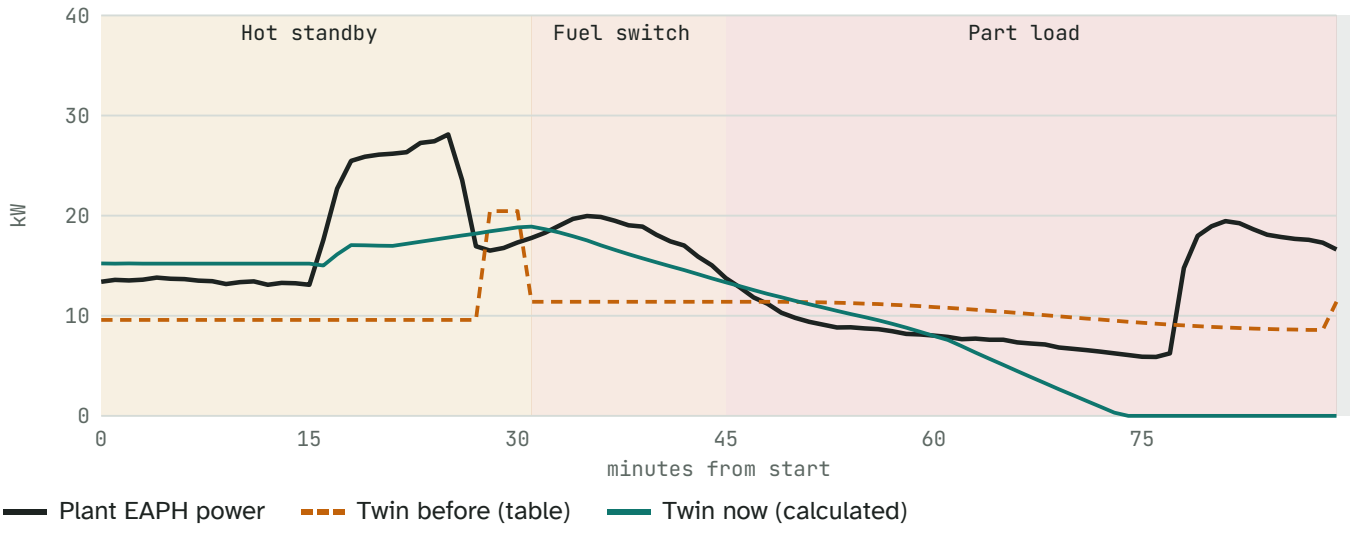
What C does: the recuperator

The air leaving the recuperators now follows the stack with a 32-minute delay, because the heat exchangers are heavy and slow. When natural gas burns it gets hotter: +0.85 °C per ln/min of gas, built up over about 32 minutes. Fitted on the 39 plant days with the 8 test periods kept aside. On those 8 periods the miss on the heater inlet dropped from 23.5 to 12.6 °C.

Heater power, before and now



Cold start, 24 h. Before: one flat number per stage (orange dashed). Now: the twin's power (green) climbs to the heat-up peak and eases down into hot standby, like the plant (black).



Hot standby → fuel switch → part load. With gas on, the power now drops as the recuperator air warms up, as on the plant. Late in part load the plant jumps back to about 18 kW while its air hardly warms across the heater. That power does not go into the air, so the twin cannot see it and reads near 0.

Score on the 8 test periods

Reading	Average miss before	Now	Inside-stage match before	Now
Outlet TIRC 2.06	22 °C	19 °C	0.96	0.99
Inlet TIR 2.05	25 °C	24 °C	0.98	1.00
Power	1.4 kW	0.7 kW	0.08	0.95

Stage	Outlet miss °C	Inlet miss °C	Power miss kW
Heat-up air	28 → 2	8 → 3	3.8 → 1.1
Heat-up N_2+H_2	18 → 1	7 → 9	3.7 → 1.5
Hot standby	2 → 0	4 → 3	1.3 → 0.6
Fuel switch	11 → 5	15 → 8	5.1 → 1.8
Part load	16 → 8	32 → 16	5.7 → 8.3
Air-flow cooling	37 → **41**	46 → **53**	0.1 → **0.1**

Green = better by more than 20 %, red = worse by more than 20 %. Air-flow cooling is the known weak spot: in the 9000 ln/min test the twin's stack itself runs about 39 °C too warm, so every reading tied to it is off too. Part-load power is the other open point. Fuel switch and part load cover only 53 and 99 minutes.

Still open

Part-load power

The plant draws 15–23 kW late in part load without warming the air. Find out where that power goes (heater losses, a second element, sensor position) before modelling it.

Air-flow cooling

The 9000 ln/min cool-down is still off by 74 °C on the stack. This is the recuperator decision from the earlier twin-vs-plant check, not new.

Long part load

With gas on for hours, the new recuperator air would pass the heater outlet and the power stays at 0. The plant never ran part load longer than 44 minutes, so this is untested.

Code: twin_2d/app.js (followRec, eaphIn, eaphPower, CALC). The stack heat balance flag is REC_PHYS = false. Backup of the old file: session scratchpad fix/app_before.js.

IF YOU REMEMBER ONE THING

The EAPH box now shows the twin's own calculation; the power rises and falls like the plant's, and only late part load stays unexplained.